

# 6-Month MLOps & AI Infrastructure Learning Path

For Web3 DevOps Engineers

Total Budget: ~\$1,400 | Time Commitment: 15-20 hours/week

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## Overview

This learning path transitions you from Web3 DevOps to MLOps and AI Infrastructure by leveraging your existing container orchestration, CI/CD, and infrastructure-as-code skills while building ML-specific competencies.

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## Month 1-2: ML Foundations & MLOps Fundamentals

Focus: Core ML concepts, Python for ML, and basic MLOps practices

Weekly Time: 15-18 hours

### Week 1-4: Machine Learning Basics

Free Resources:

- **Google Machine Learning Crash Course** (15 hours, Free)
  - URL: <https://developers.google.com/machine-learning/crash-course>
  - Focus: ML fundamentals, TensorFlow basics, model evaluation
  - Action: Complete all modules + exercises
- **Fast.ai Practical Deep Learning for Coders** (Part 1, Free)
  - URL: <https://course.fast.ai/>
  - Focus: Hands-on deep learning with PyTorch
  - Action: Complete Lessons 1-4, build your first image classifier

Supplementary:

- **NVIDIA Deep Learning Institute - Fundamentals of Deep Learning** (Free with DLI account)
  - URL: [https://learn.nvidia.com/courses/course-detail?course\\_id=course-v1:DLI+C-FX-01+V2](https://learn.nvidia.com/courses/course-detail?course_id=course-v1:DLI+C-FX-01+V2)
  - Focus: GPU-accelerated deep learning, training optimization
  - Time: 8 hours
  - Certificate: Available (adds credibility)

### Week 5-8: MLOps Foundations

Paid Investment:

- **Udacity AI Programming with Python Nanodegree** (\$399 for 3 months access)
  - URL: <https://www.udacity.com/course/ai-programming-python-nanodegree--nd089>
  - Focus: NumPy, Pandas, PyTorch, neural networks
  - Relevance: Builds coding foundation for ML
  - Time: 3 months (parallel with other learning)
  - **Start this alongside free resources to maximize value**

**Free Resources:**

- **Made With ML - MLOps Course** (Free)
  - URL: <https://madewithml.com/>
  - Focus: End-to-end ML lifecycle, experiment tracking, versioning
  - Action: Complete all MLOps modules
- **Full Stack Deep Learning** (Free, YouTube/Website)
  - URL: <https://fullstackdeeplearning.com/>
  - Focus: Production ML systems, deployment strategies
  - Time: ~12 hours of lectures

**Practice Projects (Month 1-2):**

**Project 1: Crypto Price Predictor**

- Build LSTM/Transformer model to predict BTC/ETH prices
- Data: CoinGecko API (free) + Web3.py for on-chain data
- Stack: Python, PyTorch/TensorFlow, Pandas, Docker
- Deploy: Containerize with Docker, push to Docker Hub
- MLOps: Track experiments with MLflow (free, self-hosted)
- Deliverable: GitHub repo + Docker image + prediction API (FastAPI)

**Project 2: NFT Sentiment Analysis**

- Scrape Twitter/Discord for NFT project mentions
- Train sentiment classifier (fine-tune BERT)
- Stack: Hugging Face Transformers, Tweepy, Docker
- Deploy: Docker container with model inference endpoint
- Deliverable: API that accepts text, returns sentiment score

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**Month 3: Advanced MLOps & Cloud AI Infrastructure**

**Focus:** Scalable training, model serving, cloud platforms  
**Weekly Time:** 18-20 hours

**Week 9-10: Distributed Training & GPU Management**

**Free Resources:**

- **NVIDIA DLI - Scaling Workloads Across Multiple GPUs** (Free)
  - URL: [https://learn.nvidia.com/courses/course-detail?course\\_id=course-v1:DLI+S-AC-06+V1](https://learn.nvidia.com/courses/course-detail?course_id=course-v1:DLI+S-AC-06+V1)
  - Focus: Multi-GPU training, data parallelism, NCCL
  - Time: 8 hours
- **Ray Documentation & Tutorials** (Free)
  - URL: <https://docs.ray.io/en/latest/train/train.html>
  - Focus: Ray Train for distributed ML, Ray Serve for inference
  - Action: Complete "Distributed Training" tutorial series

**Supplementary:**

- **Horovod Documentation** (Free)
  - URL: <https://horovod.readthedocs.io/>

- Focus: Distributed deep learning framework
- Action: Run example on multi-GPU setup (use AWS/GCP free tier)

## Week 11-12: Model Serving & Inference Optimization

### Paid Investment:

- **AWS Certified Machine Learning Specialty (Udemy Course)** (~\$20 on sale)
  - Instructor: Stéphane Maarek or Sundog Education
  - URL: Search "AWS Machine Learning Specialty" on Udemy
  - Focus: SageMaker, model deployment, inference endpoints
  - Time: 15 hours video + 10 hours practice

### Free Resources:

- **NVIDIA Triton Inference Server Tutorials** (Free)
  - URL: <https://github.com/triton-inference-server/tutorials>
  - Focus: High-performance model serving, multi-framework support
  - Action: Deploy a PyTorch model on Triton + Docker
- **Hugging Face Model Deployment Guide** (Free)
  - URL: [https://huggingface.co/docs/transformers/main/en/inference\\_on\\_cpu](https://huggingface.co/docs/transformers/main/en/inference_on_cpu)
  - Focus: Optimizing transformers for production
  - Action: Deploy BERT model with ONNX Runtime

## Week 13: Cloud AI Platforms Deep Dive

### Paid Investment (Choose One):

#### Option A: AWS Focus

- **AWS Free Tier** (12 months free for new accounts)
  - SageMaker: 250 hours/month of ml.t2.medium notebooks (2 months free)
  - Action: Complete SageMaker Studio tutorials, deploy 2 models

#### Option B: GCP Focus

- **Google Cloud Skills Boost - Professional ML Engineer Path** (\$50/month for 1 month)
  - URL: <https://www.cloudskillsboost.google/paths>
  - Focus: Vertex AI, BigQuery ML, TFX pipelines
  - Action: Complete 5-6 labs on Vertex AI deployment

**Recommendation:** Choose AWS (better DevOps integration for your background)

### Practice Project (Month 3):

#### Project 3: End-to-End ML Pipeline on AWS/GCP

- Problem: Predict gas fees on Ethereum network
- Data: Etherscan API + historical transaction data
- Model: XGBoost/LightGBM for regression
- Infrastructure:
  - Terraform to provision SageMaker/Vertex AI resources
  - S3/GCS for data storage
  - Lambda/Cloud Functions for data ingestion

- SageMaker/Vertex AI for training + deployment
  - MLOps:
    - GitHub Actions for CI/CD
    - Model versioning with MLflow or SageMaker Model Registry
    - Monitoring with CloudWatch/Cloud Monitoring
  - Deliverable: IaC repo (Terraform) + deployed model endpoint + monitoring dashboard
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## Month 4: Kubernetes for ML & Advanced Orchestration

**Focus: K8s operators for ML, Kubeflow, MLflow on K8s**

**Weekly Time: 16-18 hours**

### Week 14-15: Kubernetes for Machine Learning

**Paid Investment:**

- **Kubernetes for Machine Learning (Udemy)** (~\$15)
  - Search: "Kubernetes Machine Learning" or "MLOps Kubernetes"
  - Focus: K8s concepts for ML workloads, GPU scheduling
  - Time: 8-10 hours

**Free Resources:**

- **Kubeflow Documentation & Tutorials** (Free)
  - URL: <https://www.kubeflow.org/docs/started/>
  - Focus: ML workflows on Kubernetes, pipelines, serving
  - Action: Deploy Kubeflow on local Minikube or K3s cluster
  - Build a simple training pipeline (Kubeflow Pipelines)
- **NVIDIA GPU Operator Documentation** (Free)
  - URL: <https://docs.nvidia.com/datacenter/cloud-native/gpu-operator/>
  - Focus: Managing GPUs in Kubernetes clusters
  - Action: Set up GPU operator on a K8s cluster (use GKE free tier or local)

### Week 16-17: MLflow & Experiment Tracking at Scale

**Free Resources:**

- **MLflow Documentation** (Free)
  - URL: <https://mlflow.org/docs/latest/index.html>
  - Focus: Tracking, model registry, deployment
  - Action: Deploy MLflow on Kubernetes with persistent storage (PostgreSQL + S3)
- **DVC (Data Version Control) Tutorial** (Free)
  - URL: <https://dvc.org/doc/start>
  - Focus: Data versioning, pipeline orchestration
  - Action: Version control your crypto dataset with DVC + Git

### Practice Project (Month 4):

#### Project 4: Kubernetes-Based ML Platform

- Build a mini ML platform on Kubernetes:
  - Kubeflow Pipelines for training workflows

- MLflow for experiment tracking + model registry
  - Seldon Core or KServe for model serving
  - Prometheus + Grafana for monitoring
  - Use Case: Train multiple crypto prediction models (BTC, ETH, BNB) in parallel
  - Infrastructure:
    - Deploy on GKE/EKS (use free credits or local K3s)
    - Helm charts for all services
    - CI/CD with GitHub Actions (build Docker images, deploy to K8s)
  - Deliverable: GitHub repo with IaC, demo video, architecture diagram
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## Month 5: Vector Databases, LLMs & AI Pipelines

**Focus: Embeddings, vector search, LLM infrastructure**

**Weekly Time: 15-17 hours**

### Week 18-19: Vector Databases & Embeddings

**Free Resources:**

- **Pinecone Learning Center** (Free)
  - URL: <https://www.pinecone.io/learn/>
  - Focus: Vector embeddings, semantic search, RAG systems
  - Action: Complete "Vector Database Fundamentals" course
- **Weaviate Documentation & Academy** (Free)
  - URL: <https://weaviate.io/developers/academy>
  - Focus: Vector database architecture, hybrid search
  - Action: Build a semantic search app with Weaviate

**Paid Investment:**

- **Pinecone Free Tier** (Free forever for starter projects)
  - Action: Index 100K+ crypto news articles/tweets as vectors
  - Build similarity search API
- **OpenAI API Credits** (\$50 for experimentation)
  - Use for generating embeddings (text-embedding-3-small)
  - Alternative: Use free Hugging Face Inference API

### Week 20-21: LLM Infrastructure & Fine-Tuning

**Paid Investment:**

- **DeepLearning.AI - LLM Ops Specialization** (\$49/month Coursera, 1 month)
  - URL: <https://www.coursera.org/specializations/large-language-model-operations>
  - Focus: LLM deployment, fine-tuning, prompt engineering
  - Time: 3-4 weeks (accelerate to 2 weeks)
  - Certificate: Adds strong credibility

**Free Resources:**

- **Hugging Face Transformers Course** (Free)
  - URL: <https://huggingface.co/learn/nlp-course>
  - Focus: Fine-tuning LLMs, LoRA, quantization

- Action: Fine-tune LLaMA 2 or Mistral on custom dataset
- **Axolotl Fine-Tuning Framework** (Free, GitHub)
  - URL: <https://github.com/OpenAccess-AI-Collective/axolotl>
  - Focus: Simplified LLM fine-tuning with LoRA/QLoRA
  - Action: Fine-tune a 7B model on Web3 documentation

## Week 22: Advanced AI Pipelines

### Free Resources:

- **LangChain Documentation** (Free)
  - URL: [https://python.langchain.com/docs/get\\_started/introduction](https://python.langchain.com/docs/get_started/introduction)
  - Focus: Building LLM-powered applications, chains, agents
  - Action: Build a Web3 Q&A chatbot using RAG (Pinecone + OpenAI)
- **LlamaIndex Documentation** (Free)
  - URL: <https://docs.llamaindex.ai/>
  - Focus: Data connectors for LLMs, indexing strategies
  - Action: Index blockchain transaction data for LLM querying

## Practice Project (Month 5):

### Project 5: Web3-AI Hybrid Application

- **Use Case:** Smart contract-triggered AI inference via Chainlink oracles
- **Architecture:**
  - Smart contract on Ethereum testnet (Sepolia)
  - Chainlink Any API to call external AI service
  - AI service: Sentiment analysis model (Hugging Face) deployed on AWS Lambda
  - Vector database: Pinecone storing crypto news embeddings
  - Workflow:
    1. On-chain event triggers Chainlink oracle
    2. Oracle calls Lambda function with transaction data
    3. Lambda queries Pinecone for relevant context
    4. Lambda runs inference using fine-tuned model
    5. Result written back to smart contract
- **Stack:** Solidity, Hardhat, Chainlink, AWS Lambda, Pinecone, Hugging Face
- **Deliverable:** Testnet deployment + GitHub repo + demo video

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## Month 6: Advanced MLOps, Portfolio & Certification Prep

**Focus:** Production-grade MLOps, capstone project, certification

**Weekly Time:** 18-22 hours

### Week 23-24: Production MLOps Best Practices

#### Paid Investment:

- **Udacity Machine Learning DevOps Engineer Nanodegree** (\$399 for 3 months, or \$999 for 4 months with mentorship)
  - URL: <https://www.udacity.com/course/machine-learning-dev-ops-engineer-nanodegree--nd0821>
  - Focus: CI/CD for ML, model monitoring, drift detection, A/B testing

- Relevance: Directly applies DevOps skills to ML
- **Alternative:** Use remaining time from AI Programming Nanodegree if already purchased
- Certificate: Industry-recognized

## Free Resources:

- **Google MLOps Whitepaper** (Free)
  - URL: <https://cloud.google.com/resources/mlops-whitepaper>
  - Focus: MLOps maturity model, automation levels
  - Action: Assess your projects against maturity model
- **Evidently AI Documentation** (Free, open-source)
  - URL: <https://docs.evidentlyai.com/>
  - Focus: ML monitoring, data drift, model performance tracking
  - Action: Add monitoring to previous projects

## Week 25: Model Governance & Compliance

### Free Resources:

- **NVIDIA AI Enterprise Documentation** (Free)
  - URL: <https://docs.nvidia.com/ai-enterprise/>
  - Focus: Enterprise ML deployment, security, governance
  - Action: Review best practices documentation
- **MLflow Model Registry Deep Dive** (Free)
  - Focus: Model lineage, stage transitions, approval workflows
  - Action: Implement model registry with approval gates in Project 6

## Week 26: Capstone Project & Portfolio

### Capstone Project: Enterprise-Grade MLOps Platform

- **Goal:** Build a production-ready, multi-tenant ML platform
- **Features:**
  - User authentication (OAuth2)
  - Model training API (submit jobs to K8s)
  - Model registry with versioning and approval workflow
  - A/B testing framework for model deployments
  - Monitoring dashboard (data drift, performance metrics)
  - Cost tracking per model/user
- **Use Case:** Crypto trading signal generation platform
  - Multiple models (LSTM, Transformers, XGBoost) for different assets
  - Real-time inference API (<100ms latency)
  - Backtesting framework
  - Alert system (Slack/Discord webhooks)
- **Stack:**
  - Backend: FastAPI, PostgreSQL, Redis
  - ML: PyTorch, Hugging Face, MLflow
  - Infrastructure: Kubernetes (EKS), Terraform, ArgoCD
  - Monitoring: Prometheus, Grafana, Evidently AI
  - CI/CD: GitHub Actions, Docker
- **Deliverable:**
  - Production deployment (AWS/GCP)
  - Comprehensive documentation
  - 5-minute demo video

- Blog post explaining architecture
- GitHub repo with 100+ stars goal

## Week 27: Portfolio Refinement & Networking

### Free Activities:

- Polish all 6 projects on GitHub (READMEs, documentation, badges)
- Create portfolio website showcasing projects (use GitHub Pages, free)
- Write 2-3 technical blog posts (Medium, Dev.to, or personal blog)
  - Example: "From Web3 DevOps to MLOps: Lessons Learned"
  - Example: "Building a Kubernetes-Native ML Platform"
- Contribute to open-source ML projects:
  - Kubeflow, MLflow, Ray, Triton Inference Server
  - Goal: 3-5 meaningful PRs

### Networking:

- Join MLOps community Slack channels (free)
- Attend virtual MLOps meetups (meetup.com, free)
- Connect with 50+ MLOps professionals on LinkedIn
- Share your projects on Twitter/X, Reddit (r/MachineLearning, r/MLOps)

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## Curated Resource Library

### NVIDIA Training Resources (All Free with Account)

1. **Fundamentals of Deep Learning** (8 hours)
  - [https://learn.nvidia.com/courses/course-detail?course\\_id=course-v1:DLI+C-FX-01+V2](https://learn.nvidia.com/courses/course-detail?course_id=course-v1:DLI+C-FX-01+V2)
2. **Scaling Workloads Across Multiple GPUs** (8 hours)
  - [https://learn.nvidia.com/courses/course-detail?course\\_id=course-v1:DLI+S-AC-06+V1](https://learn.nvidia.com/courses/course-detail?course_id=course-v1:DLI+S-AC-06+V1)
3. **Building Transformer-Based NLP Applications** (8 hours)
  - [https://learn.nvidia.com/courses/course-detail?course\\_id=course-v1:DLI+S-FX-07+V1](https://learn.nvidia.com/courses/course-detail?course_id=course-v1:DLI+S-FX-07+V1)
4. **Deploying a Model for Inference at Production Scale** (8 hours)
  - [https://learn.nvidia.com/courses/course-detail?course\\_id=course-v1:DLI+S-FX-03+V1](https://learn.nvidia.com/courses/course-detail?course_id=course-v1:DLI+S-FX-03+V1)
5. **Building RAG Agents with LLMs** (8 hours)
  - [https://learn.nvidia.com/courses/course-detail?course\\_id=course-v1:DLI+S-FX-15+V1](https://learn.nvidia.com/courses/course-detail?course_id=course-v1:DLI+S-FX-15+V1)

**Total NVIDIA Training:** 40 hours of GPU-accelerated learning (Free)

### Udacity Resources Evaluation

#### Udacity Master of Science in AI - Infrastructure-Relevant Courses:

From the MSc AI program, these courses align with MLOps/AI Infrastructure:

1. **AI Programming Foundations** - Covered in Month 1-2 Nanodegree
2. **Machine Learning DevOps Engineer** - Recommended for Month 6
3. **Deep Learning** - Useful but expensive (~\$1,800 for full degree)

**Recommendation:** Skip the full MSc AI degree (\$9,600 total). Instead, take the two Nanodegrees (\$399 each = \$798) which provide 80% of the value for 8% of the cost. The MSc is better for those needing a formal degree for visa/career



requirements.

## Free Alternative Resources

### If Budget is Tight (Sub-\$500 Path):

- **Replace Udacity Nanodegrees with:**
  - Google Machine Learning Crash Course (Free)
  - Fast.ai (Free)
  - Made With ML (Free)
  - Full Stack Deep Learning (Free)
- **Replace Coursera LLMOps with:**
  - Hugging Face Transformers Course (Free)
  - DeepLearning.AI short courses on YouTube (Free)
- **Replace Cloud Courses with:**
  - AWS/GCP free tier documentation (Free)
  - YouTube tutorials from Tech With Tim, Krish Naik (Free)

**Total Low-Budget Path:** <\$150 (just Udemy courses + API credits)

## Budget Breakdown

### Paid Resources Summary

| Resource                                      | Cost  | Month | Notes                           |
|-----------------------------------------------|-------|-------|---------------------------------|
| Udacity AI Programming with Python Nanodegree | \$399 | 1-3   | 3-month access                  |
| Udacity ML DevOps Engineer Nanodegree         | \$399 | 6     | Alternative: extend previous ND |
| AWS ML Specialty Course (Udemy)               | \$20  | 3     | Wait for sale                   |
| Kubernetes for ML (Udemy)                     | \$15  | 4     | Wait for sale                   |
| DeepLearning.AI LLMOps (Coursera)             | \$49  | 5     | 1 month subscription            |
| Google Cloud Skills Boost (Optional)          | \$50  | 3     | Alternative to AWS              |
| OpenAI API Credits                            | \$50  | 5     | For embeddings + testing        |
| AWS/GCP Cloud Costs                           | \$150 | 3-6   | SageMaker, EKS, Vertex AI       |
| Domain + Hosting (Portfolio)                  | \$20  | 6     | Namecheap + Netlify Pro         |
| Contingency (Books, Tools)                    | \$50  | 1-6   | O'Reilly, Kaggle competitions   |

**Total: \$1,202** (within \$1,400 budget, \$198 buffer)

### Free Resources Value (If Purchased)

| Resource                       | Equivalent Value |
|--------------------------------|------------------|
| NVIDIA DLI Courses (5 courses) | \$3,000+         |
| Fast.ai Course                 | \$1,000+         |
| Google ML Crash Course         | \$500+           |
| Made With ML                   | \$800+           |
| Hugging Face NLP Course        | \$600+           |
| Kubeflow/MLflow Documentation  | \$1,200+         |
| Open-source tools practice     | \$2,000+         |

**Total Free Value: \$9,100+**

# Cost Optimization Tips

1. **Udemy Sales:** Never pay full price. Courses go on sale for \$10-20 every 2 weeks
  2. **Cloud Free Tiers:**
    - AWS: 12 months free tier (SageMaker, EC2, S3)
    - GCP: \$300 free credits for 90 days
    - Azure: \$200 free credits for 30 days
  3. **Student Discounts:**
    - GitHub Student Pack: Free cloud credits, tools
    - Coursera Financial Aid: 100% off if approved (4-week process)
  4. **NVIDIA Credits:** Free GPUs via their LaunchPad program (application required)
  5. **Kaggle Notebooks:** Free GPU/TPU access (30 hours/week)
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## Learning Path Milestones & Success Metrics

### Month 1-2 Milestones:

-  Complete 3 Google/NVIDIA free courses
-  Build 2 Dockerized ML projects
-  Track 10+ experiments in MLflow
-  Achieve 75%+ accuracy on crypto price predictor
-  Deploy 1 model API to cloud

### Month 3 Milestones:

-  Complete AWS ML course
-  Deploy 1 model on SageMaker with Terraform
-  Set up distributed training with Ray
-  Implement model monitoring
-  Document end-to-end pipeline

### Month 4 Milestones:

-  Deploy Kubeflow on Kubernetes cluster
-  Build 3 Kubeflow pipelines
-  Run multi-GPU training job
-  Set up MLflow on K8s with PostgreSQL backend
-  Implement CI/CD for ML models

### Month 5 Milestones:

-  Index 100K+ vectors in Pinecone
-  Fine-tune 1 LLM (7B+ parameters)
-  Build Web3-AI hybrid app with Chainlink
-  Deploy RAG system
-  Complete LLMOps specialization

## Month 6 Milestones:

-  Launch capstone project to production
-  Publish 3 technical blog posts
-  Contribute 5 PRs to open-source ML projects
-  Portfolio website live with 6 projects
-  Complete ML DevOps Nanodegree
-  Apply to 10+ MLOps/AI Infrastructure roles

## Success Criteria (End of Month 6):

### Technical Skills:

- Train, tune, and deploy ML models end-to-end
- Manage Kubernetes clusters with GPU workloads
- Build CI/CD pipelines for ML (training + inference)
- Monitor models in production (drift, performance)
- Design scalable AI infrastructure with IaC
- Work with vector databases and LLMs

### Portfolio:

- 6 production-quality projects on GitHub
- 1 capstone project deployed to cloud
- 3+ technical blog posts
- Active open-source contributions
- Portfolio website showcasing work

### Certifications (Optional but Recommended):

- Udacity ML DevOps Engineer Nanodegree
- 5 NVIDIA DLI Certificates
- AWS ML Specialty (if pursuing certification exam)

### Network:

- 100+ LinkedIn connections in MLOps/AI
- Active in 3+ MLOps communities
- 2-3 informational interviews with MLOps engineers

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## Recommended Tools & Setup

### Development Environment

#### Local Setup:

- **Laptop:** 16GB+ RAM, GPU preferred (RTX 3060+ or M1/M2 Mac)
- **OS:** Linux (Ubuntu 22.04) or WSL2 on Windows
- **Python:** 3.10+ with pyenv
- **IDE:** VS Code with extensions (Python, Docker, Kubernetes, Jupyter)
- **Docker Desktop:** Latest version
- **kubect**l + **k9s:** Kubernetes CLI tools

- **Terraform:** Latest version
- **Git + GitHub CLI**

**Cloud Accounts:**

- AWS Free Tier (12 months)
- GCP Free Trial (\$300 credits)
- Azure Free Trial (\$200 credits)
- Kaggle (free GPU/TPU notebooks)

**ML Tools:**

- **Frameworks:** PyTorch, TensorFlow, Scikit-learn
- **MLOps:** MLflow, DVC, Kubeflow
- **Monitoring:** Evidently AI, Prometheus, Grafana
- **Vector DBs:** Pinecone (free tier), Weaviate
- **LLM:** Hugging Face Transformers, LangChain, LlamaIndex

**Suggested Hardware Investment (Optional)**

If you plan to train models locally (beyond cloud free tiers):

- **Budget GPU Server** (~\$1,500-2,000):
  - Used RTX 3090 (24GB VRAM): ~\$800
  - Ryzen 9 5900X + Motherboard + RAM: ~\$700
  - 2TB NVMe SSD: ~\$150
  - PSU + Case: ~\$200

**ROI:** Saves \$30-50/month on cloud GPU costs, pays for itself in 1-2 years

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 **Career Transition Strategy**

**Resume Optimization**

**Headline Example:** "DevOps Engineer | Web3 Infrastructure → Transitioning to MLOps & AI Infrastructure | 6 Production ML Projects | AWS | Kubernetes | PyTorch"

**Key Skills to Highlight:**

- ML model deployment (SageMaker, Vertex AI, Kubernetes)
- CI/CD for ML (GitHub Actions, Jenkins, MLflow)
- Infrastructure as Code (Terraform, Helm, ArgoCD)
- Container orchestration (Docker, Kubernetes, GPU scheduling)
- Model monitoring & observability (Prometheus, Grafana, Evidently)
- Distributed training (Ray, Horovod, multi-GPU)
- Vector databases & LLM infrastructure (Pinecone, Triton)

**Job Search Timeline**

**Month 3-4:** Start informational interviews with MLOps engineers **Month 5:** Begin applying to junior/mid-level MLOps roles **Month 6:** Intensive job search (10-15 applications/week)

# Target Roles

## Entry-Level Targets:

- ML Infrastructure Engineer (Junior)
- MLOps Engineer (1-2 years exp)
- DevOps Engineer (ML/AI teams)
- AI Platform Engineer

## Companies to Target:

- AI startups (high growth, more willing to train)
- Web3 companies building AI products (leverage your background)
- Cloud providers (AWS, GCP, Azure ML teams)
- MLOps tooling companies (Weights & Biases, Comet, Tecton)

## Salary Expectations (US, 2025):

- Junior MLOps Engineer: \$90K-130K
- Mid-Level MLOps Engineer: \$130K-180K
- Senior MLOps Engineer: \$180K-250K+

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## Weekly Schedule Template

### Example Week (Months 3-5)

#### Monday (4 hours):

- 2 hours: Course videos/reading
- 2 hours: Project work

#### Tuesday (3 hours):

- 1 hour: NVIDIA DLI course
- 2 hours: Hands-on lab/tutorial

#### Wednesday (4 hours):

- 3 hours: Project implementation
- 1 hour: Blog post writing

#### Thursday (3 hours):

- 2 hours: Course completion
- 1 hour: Open-source contribution

#### Friday (2 hours):

- 2 hours: Project testing/debugging

#### Saturday (5 hours):

- 3 hours: Deep dive project work

- 2 hours: Documentation

### Sunday (2 hours):

- 1 hour: Weekly review & planning
- 1 hour: Networking (LinkedIn, Slack communities)

**Total: 23 hours/week** (adjust to your availability)

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## Optional Certifications (Beyond 6 Months)

If you want to continue learning and add certifications:

1. **AWS Certified Machine Learning – Specialty** (~\$300 exam)
    - Prep time: 2-3 months
    - Value: High (recognized by employers)
  2. **Google Professional ML Engineer** (~\$200 exam)
    - Prep time: 2-3 months
    - Value: High (especially for GCP roles)
  3. **TensorFlow Developer Certificate** (~\$100 exam)
    - Prep time: 1 month
    - Value: Medium (validates TensorFlow skills)
  4. **Certified Kubernetes Administrator (CKA)** (~\$395 exam)
    - Prep time: 2 months
    - Value: High (complements MLOps skills)
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## Support & Community Resources

### Communities to Join (All Free)

1. **MLOps Community** (Slack)
  - URL: <https://mlops.community/>
  - 20K+ members, job postings, events
2. **Hugging Face Discord**
  - URL: <https://discuss.huggingface.co/>
  - LLM discussions, model sharing
3. **r/MachineLearning** (Reddit)
  - URL: <https://www.reddit.com/r/MachineLearning/>
  - Research papers, discussions
4. **r/MLOps** (Reddit)
  - URL: <https://www.reddit.com/r/mlops/>
  - Best practices, tool discussions
5. **NVIDIA Developer Forums**
  - URL: <https://forums.developer.nvidia.com/>
  - GPU, CUDA, Triton support
6. **Made With ML Slack**
  - URL: <https://madewithml.com/>
  - Active learners, project feedback

## Mentorship

- **NVIDIA Inception Program** (Free for startups)
  - Technical mentorship + cloud credits
- **MLOps Community Mentorship Program** (Free)
  - Pair with experienced MLOps engineers

## Final Tips for Success

### Do's:

- ✓ Build in public (share progress on LinkedIn/Twitter)
- ✓ Focus on end-to-end projects (not just notebooks)
- ✓ Leverage your DevOps expertise (IaC, CI/CD, K8s)
- ✓ Network actively (50% learning, 50% networking)
- ✓ Contribute to open source (builds credibility)
- ✓ Document everything (blog posts, READMEs)
- ✓ Stay consistent (small daily progress > weekend binges)

### Don'ts:

- ✗ Don't try to learn everything (focus on depth in MLOps)
- ✗ Don't skip projects (hands-on practice is 80% of learning)
- ✗ Don't work in isolation (join communities early)
- ✗ Don't pay for redundant courses (free resources are abundant)
- ✗ Don't wait until "ready" to apply (start at Month 5)
- ✗ Don't neglect soft skills (communication, collaboration)

### Mindset:

- **Imposter syndrome is normal** - everyone feels it transitioning fields
- **Leverage your DevOps experience** - it's a huge advantage in MLOps
- **Web3 + AI is a unique niche** - position yourself as a specialist
- **Fail fast, iterate** - most learning happens through debugging projects
- **Quality > Quantity** - 6 great projects > 20 mediocre ones

## Progress Tracking Dashboard

Create a simple spreadsheet to track:

| Week | Hours | Logged Courses  | Completed Projects | Progress | Blog Posts | PRs Submitted | Connections Made |
|------|-------|-----------------|--------------------|----------|------------|---------------|------------------|
| 1    | 18    | ML Crash (50%)  | Project 1 (20%)    | 0        | 0          |               | 5                |
| 2    | 20    | ML Crash (100%) | Project 1 (60%)    | 1        | 0          |               | 10               |
| ...  | ...   | ...             | ...                | ...      | ...        |               | ...              |

### Monthly Review Questions:

1. Did I hit my hour goals? (Why or why not?)

2. Are my projects deployment-ready? (Not just notebooks)
  3. Am I actively networking? (10+ new connections/month)
  4. What did I struggle with? (Adjust next month's focus)
  5. What was my biggest win? (Celebrate it!)
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## Month 6 Graduation Checklist

By the end of Month 6, you should have:

- ☐ 6 production-ready projects on GitHub (all with READMEs, Docker, CI/CD)
- ☐ 1 capstone project deployed to cloud (with live demo)
- ☐ Portfolio website with case studies
- ☐ 3+ published blog posts (Medium/Dev.to)
- ☐ 2 Udacity Nanodegree certificates
- ☐ 5 NVIDIA DLI certificates
- ☐ 5+ open-source contributions
- ☐ 100+ LinkedIn connections in MLOps/AI
- ☐ Resume updated with MLOps skills
- ☐ Applied to 20+ MLOps roles
- ☐ 3+ informational interviews completed

**Congratulations! You're now an MLOps Engineer!** 🎓🚀

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## Next Steps

After completing this 6-month path:

1. **Immediate Job Search:** Apply to 50+ roles over 2 months
2. **Keep Learning:** ML evolves fast - stay current with papers, tools
3. **Consider Specialization:**
  - LLM Infrastructure (hot market in 2025)
  - Computer Vision MLOps
  - Edge AI / Model Optimization
  - ML Platform Engineering
4. **Give Back:** Mentor others transitioning into MLOps
5. **Build a Personal Brand:** Keep blogging, speaking, contributing

**Good luck on your journey from Web3 DevOps to MLOps!** 🚀

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*Last Updated: October 2025*

*Feel free to adapt this plan to your schedule and learning preferences!*